

- 5 (a) (i) magnetic flux density decreases with distance from straight wire, so magnetic flux linkage in loop decreases as it falls M1
- e.m.f. induced in loop proportional to rate of change of magnetic flux linkage A1
- OR
- loop cuts magnetic flux as it falls (M1)
- e.m.f. induced proportional to rate of cutting of magnetic flux (A1)
- (ii) magnetic flux into page through the loop decreasing B1
- induced current produces magnetic flux into page to oppose the decrease B1
- (right hand grip rule gives) induced current flowing clockwise in loop B1
- OR
- magnetic flux density is higher in upper half than lower half of loop, so rate of flux cut and induced e.m.f. larger in upper half than lower half of loop (B1)
- direction of induced e.m.f. (drives induced current left to right) is same in both halves, so net e.m.f. (B1)
- drives current clockwise in loop (B1)
- (b) induced e.m.f.
- $$= -\frac{\Delta\Phi}{\Delta t} = -\frac{(30-120)\times 10^{-3}\times \pi \times 0.050^2}{0.040}$$
- M1
- $$= 0.0177 \text{ V} = 18 \text{ mV}$$
- A0

